

Diabetes Prediction and Factor Analysis using Logistic Regression

Introduction

Diabetes is a chronic medical condition that affects a large population worldwide. Early prediction and identification of risk factors can help in better prevention and treatment. This project focuses on building a machine learning model to predict whether an individual is diabetic and to analyze the impact of various health and lifestyle factors.

Objective

- To develop a classification model to predict diabetes.
- To evaluate the performance of the model using standard metrics.
- To identify and analyze the key factors influencing diabetes.

Dataset Description

The dataset contains patient-related information such as:

- Gender
- Age
- Hypertension Blood glucose levels
- Heart disease
- Smoking history
- BMI
- HbA1c_level
- blood_glucose_level

The target variable:

- Diabetes (0 = No, 1 = Yes)

Data Preprocessing

- Categorical variables such as *gender* and *smoking history* were converted into numerical format using one-hot encoding.
- The `drop_first=True` parameter was used to avoid multicollinearity by removing one category as a baseline.
- The dataset was divided into: Features (X) and Target variable (y)

Model Used

A **Logistic Regression** model was used for classification.

Reason for choosing Logistic Regression:

- Suitable for binary classification problems
- Easy to interpret
- Provides insight into feature impact through coefficients

Model Training

- The dataset was split into training and testing sets.
- The model was trained using the training data.
- Maximum iterations were increased to ensure proper convergence.

Model Evaluation

Accuracy

The model achieved high accuracy, indicating strong predictive performance.

Confusion Matrix

The confusion matrix shows

- Correct predictions of diabetic and non-diabetic cases
- False positives (incorrectly predicted diabetic)
- False negatives (missed diabetic cases)

This helps in understanding model reliability, especially in identifying actual diabetic patients.

Classification Report

- **Precision:** Accuracy of positive predictions
- **Recall:** Ability to detect actual diabetic cases
- **F1-score:** Balance between precision and recall

Recall is particularly important in this context, as missing a diabetic case can have serious consequences.

Feature Importance Analysis

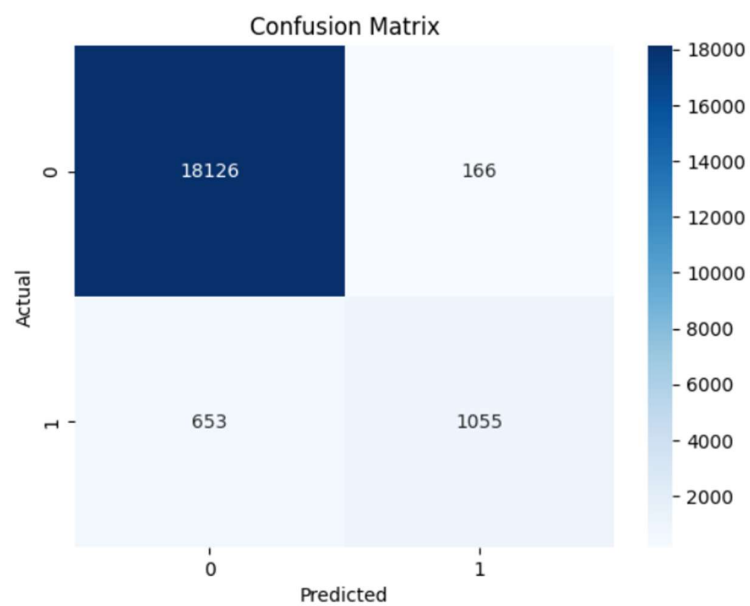
- Model coefficients were used to determine the impact of each feature.
- Positive values indicate higher likelihood of diabetes.
- Negative values indicate lower likelihood.

Key Insights:

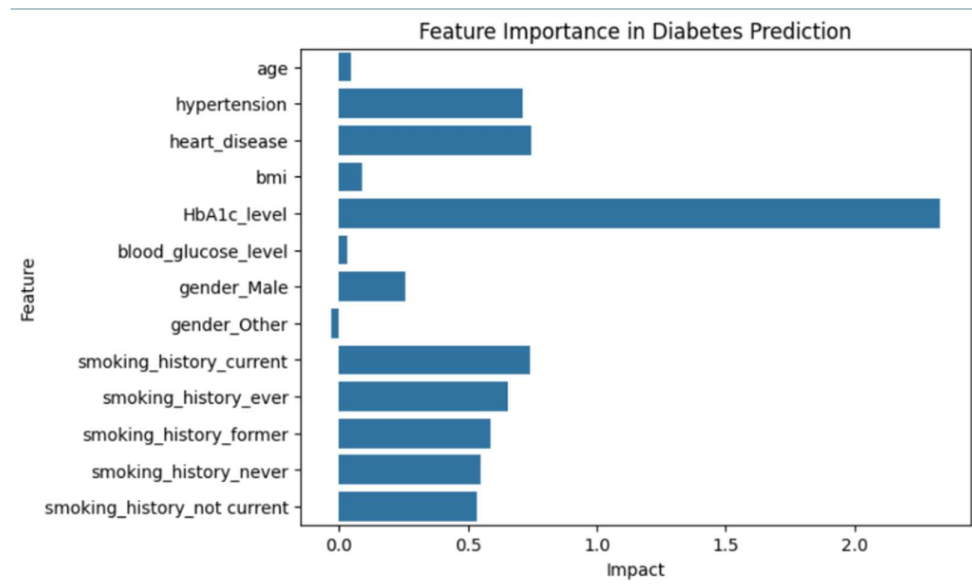
- Certain features such as BMI, glucose levels, and lifestyle factors significantly influence diabetes.
- The model helps identify high-risk individuals based on these factors.

Visualization

- Confusion matrix was visualized using a heatmap for better interpretation.



- Feature importance was displayed using a bar chart to highlight impactful variables.



Conclusion

The Logistic Regression model effectively predicts diabetes with high accuracy. The analysis also highlights important factors contributing to diabetes, which can be useful for early detection and preventive healthcare strategies.

Tools and Libraries Used

- Python
- Pandas (Data preprocessing)
- NumPy
- Scikit-learn (Model building)
- Matplotlib & Seaborn (Visualization)